



**PLATING POSTERS**

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PLATING POSTERS INC • METAL FINISHING REFERENCE SERIES

# IRON PHOSPHATE

## THE COMPLETE TRAINING MANUAL

CONVERSION COATING • PAINT BASE

*A full training course for the brand-new operator — from “what even is a conversion coating?” to a signed certificate of completion. Light amorphous iron phosphate on steel — the workhorse pretreatment under paint and powder coat. No electricity, no plating tanks, no mystery. Assumes you know nothing. Teaches you everything.*



### OPERATOR TRAINING & CERTIFICATION

EDITION 1.0 • 2026 • FOR-DUMMIES STYLE

Training reference only. Always verify exact parameters against your chemistry supplier's Technical Data Sheets (TDS), customer specifications, current Safety Data Sheets (SDS), and applicable OSHA, EPA, REACH, and local regulations before production use.

— FIND YOUR WAY

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READ FRONT-TO-BACK THE FIRST TIME · KEEP ON THE SHELF FOREVER AFTER

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### REMEMBER

This manual is a training and reference tool — **not** a substitute for your facility's written procedures, your supplier's TDS/SDS, or your supervisor's instructions. When this manual and your shop's official procedure disagree, **follow your shop's official procedure and ask your supervisor.**

# WELCOME ABOARD

## IF YOU DON'T KNOW A CONVERSION COATING FROM A COFFEE CUP, YOU ARE IN EXACTLY THE RIGHT PLACE

Welcome. If you're holding this manual, you're about to learn how to run — or at least understand — an **iron phosphate line**: the chemical pretreatment that goes on steel parts *before* they get painted or powder coated. Maybe you started this week. Maybe you've been hanging parts on the conveyor for a year and finally want to know *why* you do the things you do. Either way, you're in the right place, and we're glad you're here.

Here's the most important thing we can tell you up front: **this manual assumes you know nothing about chemistry, metal finishing, or how paint sticks to steel.** That's not an insult — it's a promise. Every term gets defined the first time we use it. Every step explains not just *what* to do but *why* it matters. Nobody is born knowing what “coating weight” means or why a tired rinse tank can quietly ruin a whole paint job. You learn it. We're going to teach it — in the spirit of the *For Dummies* books: friendly, plain-spoken, and patient.



### REMEMBER

You do not need to memorize this manual. You need to *understand* it. Understanding sticks. Memorizing fades by Friday.



### HEADS-UP

One big idea before we start: **an iron phosphate line has no electricity running through the parts.** There is no rectifier, no anode, no cathode — the coating forms because the steel and the chemistry *react with each other*. That removes the shock-at-the-part hazard, but it does **not** make the line safe to ignore: you're still working with hot caustic cleaners, mild acid, oxidizing accelerators, possibly chromium seal chemistry, hot ovens, slick floors, and walk-in washer tunnels. Respect all of it.

## • HOW TO USE THIS MANUAL

The manual follows the **line itself**, station by station, in the order a part actually travels: inspect it, clean it, rinse it, phosphate it, rinse it, seal it, dry it, and hand it to paint. Reading it in order matters, because (as you'll soon learn) **the order of a pretreatment line is not random** — and neither is the order of this book.

## THE ICONS — YOUR FRIENDLY MARGIN HELPERS

As you read, you'll run into little flagged notes. Here's what each one means.



### TIP

A shortcut or trick of the trade — the kind of thing a 20-year veteran would lean over and tell you.



### REMEMBER

A core idea worth locking in. If you forget everything else on the page, don't forget these — they're the load-bearing ideas.



### HEADS-UP

A safety warning. Read every one — these keep your skin, eyes, and lungs intact. We use them *liberally*, because a pretreatment line has real hazards.



### TECHNICAL STUFF

Deeper chemistry for the curious. Totally optional — skip it and you can still run the line perfectly.



### FUN FACT

A bit of trivia to make the science stick (and give you something to say at lunch).

## TWO THINGS YOU'LL SEE AT EVERY STATION

Each station chapter ends with two recurring tools. The **Teach-Back** checklist (often paired with a red “Top Mistakes” card) is the set of things you must be able to *say out loud, unprompted* before you're cleared. The bordered **Sign-Off** box is the short statement your *trainer initials* once you've demonstrated the skill safely. Teach-Back is your study guide; Sign-Off is the gate.

WHAT YOU'LL BE ABLE TO DO

# LEARNING OBJECTIVES

## MASTER THIS FOUNDATION AND EVERY STATION CHAPTER CLICKS INTO PLACE

By the time you finish this manual, you should be able to confidently:

1. **Explain what a conversion coating is** in plain language — and how it differs from electroplating and anodizing.
2. **Describe how iron phosphate forms** — a chemical reaction between the steel surface and the solution, with *no electric current* — and what the coating is actually for (a paint/powder base).
3. **Name all eight stations of the line in order** and explain *why* the order cannot be rearranged.
4. **Read coating weight and color** — know that iron phosphate runs light (~30–90 mg/ft<sup>2</sup>), that color goes blue→gold→tan with increasing weight, and what “too light” and “too heavy” look like.
5. **Explain the cleaner-coater concept** — how one product can both clean and coat, and why that lets some lines run in just three stages.
6. **Recognize the main hazards** at each station — hot caustic, mild acid, oxidizing accelerators, chromium seal chemistry, hot ovens, mist, slips, and washer-tunnel confined spaces — and know the correct PPE and response for each.
7. **Use the basic field checks** every operator relies on — the water-break test, coating color, and simple coating-weight verification.
8. **Spot trouble early** — read a part and know whether the problem is upstream (cleaning), in the phosphate stage, in the rinse, or in the seal/dry.
9. **Talk the language** — use the right words so you can communicate clearly with your lead, your chemist, and your chemistry supplier.



### REMEMBER

You are not expected to hit all of these on day one. Surface finishing is a craft. The goal is steady competence, not instant mastery.

## • THE EIGHT-STATION PROCESS AT A GLANCE

Here's the whole iron phosphate line in one strip. Don't memorize it yet — just get a feel for the rhythm: **Inspect** → **Clean** → **Rinse** → **Phosphate** → **Rinse** → **Seal** → **Dry-Off** → **Inspect/Paint**.



### REMEMBER

Notice the pattern: **a rinse follows almost every working stage**. Rinses aren't breaks — they're checkpoints that keep one chemistry from poisoning the next. Every working solution wants to “hitch a ride” into the next stage on the surface of your part, and rinses are what stop that. They aren't filler between the “real” steps; they're some of the most important steps you'll perform.



### HEADS-UP

This eight-station layout is the *teaching* version — the classic **5-stage spray washer** (clean / rinse / phosphate / rinse / seal) bookended by inspect, dry-off, and the paint hand-off. Your shop may run a leaner **3-stage** line (cleaner-coater / rinse / seal). Both are covered. Always run *your* line's actual sequence and posted parameters.

## CHAPTER 1

# WHAT IS A CONVERSION COATING?

THE TRUE BEGINNER'S PRIMER — SO FAR BACK WE DON'T EVEN ASSUME YOU KNOW WHAT "FINISHING" MEANS

## THE ONE-SENTENCE VERSION

A conversion coating is a protective film you grow on a metal surface by making the metal itself react with a chemical solution — turning the top layer of the metal *into* the coating. That's the whole idea. Now let's unpack it, because the word "conversion" is doing a lot of work.

## THREE COUSINS: PLATING, ANODIZING, AND CONVERSION COATING

If you've seen our other manuals, you've met two other finishing families. It's worth lining all three up side by side, because the differences are the key to understanding iron phosphate.

- **Electroplating** *adds* a layer of a *different* metal on top of your part, using **electricity**. Electricity drives dissolved metal (like zinc) out of a bath and onto the part. The coating is a brand-new metal skin sitting *on* the original surface.
- **Anodizing** *converts* the part's *own* surface into a thick, hard oxide — but it still uses **electricity** to drive the reaction. It only works on metals like aluminum. The coating is grown *out of* the part itself, electrochemically.
- **Conversion coating** (this manual) *converts* the part's own surface into a thin protective film using **only chemistry — no electricity at all**. You dunk or spray the part, the surface reacts with the solution, and a new compound (here, iron phosphate plus iron oxides) forms right where the bare metal used to be.



### REMEMBER

Lock in the difference: **Plating adds a new metal with electricity. Anodizing converts the surface with electricity. A conversion coating converts the surface with chemistry alone — no rectifier, no anode, no cathode, no current.** If you ever go looking for the "power supply" on an iron phosphate line, you won't find one feeding the parts. That's not a missing piece — that's the whole point.



### FUN FACT

Iron phosphating took off in the 1930s–40s as the auto and appliance industries hunted for a fast, cheap way to make paint stop peeling off steel. The breakthrough was realizing you could *clean and coat in one step* with the right chemistry — the ancestor of today's "cleaner-coater." Your modern spray washer is the great-grandchild of those first dunk tanks.

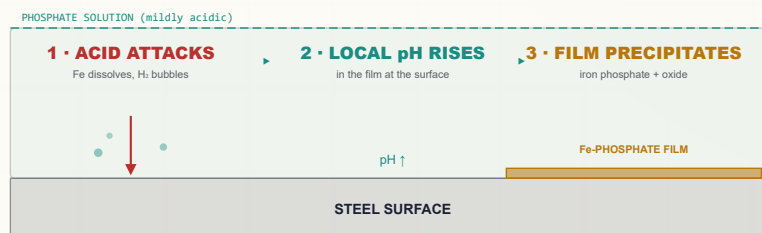
## HOW A CONVERSION COATING FORMS — NO CURRENT REQUIRED

So if there's no electricity pushing metal around, what makes the coating appear? The metal and the solution do it themselves. Here's the plain-language version for iron phosphate:

1. The phosphate solution is **mildly acidic** (it contains phosphoric acid). When a clean steel part touches it, the acid begins to gently dissolve — “pickle” — the very top atoms of the steel surface. Iron goes into solution, and tiny hydrogen bubbles form. (You'll sometimes see fine bubbling on parts in an immersion tank — that's the reaction working.)
2. Right at the metal surface, that pickling reaction uses up acid, so the solution there becomes **less acidic** (the local pH climbs) in a microscopically thin film hugging the part.
3. When the local acidity drops past a tipping point, the dissolved iron and the phosphate in the solution can no longer stay dissolved at the surface — they **precipitate**, depositing as a thin, tightly-bonded film of **iron phosphate and iron oxide** right onto the steel.
4. An **accelerator** in the bath (an oxidizer) keeps the reaction moving briskly and helps build a uniform film instead of a slow, patchy one.

The result is an **integral** coating — it's chemically grown into the surface, not glued on top — and it's exactly the kind of slightly rough, chemically receptive surface that paint and powder love to grab onto.

### HOW THE FILM FORMS (NO ELECTRICAL CELL)



*No electricity — the metal and the solution do it themselves.*



#### TECHNICAL STUFF

Roughly: the acid attacks iron, e.g.  $\text{Fe} + 2 \text{H}_3\text{PO}_4 \rightarrow \text{Fe}(\text{H}_2\text{PO}_4)_2 + \text{H}_2\uparrow$ . That consumes free acid and raises the interfacial pH, which drives precipitation of the amorphous film — a mix of iron phosphate (often written  $\text{FePO}_4$ ) and iron oxides ( $\text{Fe}_3\text{O}_4/\text{Fe}_2\text{O}_3$ ). Despite the name, an “iron phosphate” coating is *not* pure phosphate — it's a blend, and it's **amorphous** (no regular crystal structure), which is part of why it stays so thin and light. Its cousins zinc phosphate and manganese phosphate form *crystalline*, much heavier coatings.



#### HEADS-UP

“No electricity at the part” removes the shock-at-the-tank hazard you'd have on a plating line — but pumps, conveyors, heaters, and oven controls on a washer are still powered by serious electrical equipment. **Any maintenance still requires Lockout/Tagout (LOTO).** No current through the parts does not mean no current in the building.

CHAPTER 2

# WHAT IS IRON PHOSPHATE?

THE LIGHT, AMORPHOUS COATING WHOSE ONE JOB IS TO MAKE PAINT STICK — AND STAY STUCK

## • IRON PHOSPHATE’S ONE BIG JOB: A PAINT & POWDER BASE

Here’s the headline, and it’s the single most important idea in this whole manual: **iron phosphate is a pretreatment. Its job is to prepare steel so that paint or powder coat sticks well and resists corrosion creeping under the paint.** It is **not** a standalone, heavy-duty rust coating you leave bare. On its own, an unpainted iron phosphate coating offers only modest, short-term corrosion protection. Painted over, it transforms the finish — dramatically better adhesion and dramatically better resistance to the paint blistering, peeling, or rusting from underneath.

Why does such a thin film help so much? Two reasons:

1. **Mechanical “tooth.”** The conversion process leaves a finely textured, slightly roughened surface with far more surface area than bare, smooth steel. Paint and powder grip that texture like Velcro instead of sliding on glass.
2. **A buffer against under-paint corrosion.** Bare steel under paint is a launch pad for rust — once moisture finds a scratch or a pinhole, corrosion races *along the steel/paint boundary* and lifts the coating. The iron phosphate film converts that reactive bare steel into a stable, inert surface, slowing that under-film creep way down.



### REMEMBER

**Iron phosphate is a *paint base*, not a *finish*.** Nobody ships bare iron-phosphated parts and calls them done. The phosphate is step one of two: phosphate, then paint/powder. If anyone treats the phosphate as the final corrosion coating, that’s a misunderstanding worth correcting.

## • COATING WEIGHT & COLOR — HOW WE “SEE” THE COATING

Because the coating is so thin, we measure it not in thickness but in **coating weight** — how many milligrams of coating sit on each square foot (or each square meter) of surface.

- **Typical iron phosphate:** ~30–90 mg/ft² (~0.3–1.0 g/m²). That’s *light* — a fraction of a thousandth of an inch.
- For comparison, **zinc phosphate** runs roughly 150–1,000+ mg/ft² (1.5–10+ g/m²) and **manganese phosphate** heavier still — those are different products we’ll cover in their own manuals.

The wonderful thing about iron phosphate is that **you can read its weight by its color.** As the coating gets heavier, light bouncing off the thin film shifts color (the same physics that makes a soap bubble or an oil slick shimmer):

| COLOR YOU SEE                  | ROUGHLY WHAT IT MEANS            | VERDICT FOR PAINT BASE                    |
|--------------------------------|----------------------------------|-------------------------------------------|
| No color / gray (bare-looking) | Little to no coating formed      | Too light — poor adhesion                 |
| Faint blue / blue-iridescent   | Light coating (~30–50 mg/ft²)    | Good for many jobs                        |
| Gold / golden iridescent       | Medium coating (~50–80 mg/ft²)   | Strong paint base — the sweet spot        |
| Tan / brown, powdery or dusty  | Over-heavy coating (>~90 mg/ft²) | Too heavy — loose, chalky, worse adhesion |



### TIP

Operators run iron phosphate “by color” every day. A consistent **blue-to-gold iridescence** across the part is your fast, free, on-the-line confirmation the process is in the zone. **Tan, brown, or a wipe-off powder is a red flag:** the coating is too heavy and will actually hurt paint adhesion, not help it. Heavier is *not* better with iron phosphate.

★ **REMEMBER**

Two failure directions, not one. **Too light** (no color, bare gray) = not enough coating = paint won't grip. **Too heavy** (tan/powdery) = a loose, chalky layer that paint grips *to*, then peels off *with*. The target is the iridescent blue-to-gold middle. Always confirm the exact target weight/color on your job spec.

READING COATING WEIGHT BY COLOR • AND WHERE THE FILM SITS

TARGET ~30–90 mg/ft<sup>2</sup>

GRAY  
too light

BLUE

GOLD

TAN  
too heavy

PAINT / POWDER

SEAL (thin)

IRON PHOSPHATE FILM (thin, integral)

STEEL SUBSTRATE

← paint grips here

The phosphate is the grip layer between the steel and the paint.

• WHERE IRON PHOSPHATE FITS IN THE PHOSPHATE FAMILY

Phosphating is a whole family. Knowing where iron phosphate sits keeps you from using the wrong tool:

| COATING                      | COATING WEIGHT (TYPICAL)        | CRYSTAL/FEEL         | COLOR              | MAIN JOB                                   |
|------------------------------|---------------------------------|----------------------|--------------------|--------------------------------------------|
| Iron phosphate (this manual) | 30–90 mg/ft <sup>2</sup>        | Amorphous, very thin | Blue→gold          | Paint/powder base                          |
| Zinc phosphate (next manual) | 150–1,000+ mg/ft <sup>2</sup>   | Crystalline, heavier | Gray               | Heavier paint base + corrosion + lube base |
| Manganese phosphate          | 1,000–2,000+ mg/ft <sup>2</sup> | Crystalline, heavy   | Dark gray to black | Wear, break-in, oil retention              |

⚙️ **TECHNICAL STUFF**

Iron phosphate is the lightest, cheapest, and simplest of the three, and the easiest to run at low temperature. It's the right answer when the goal is "good paint adhesion on steel at low cost." When the customer needs *more* under-paint corrosion resistance they step up to **zinc phosphate**. When they need a thick, oil-holding, wear-resistant black coating for sliding/mating steel parts, they use **manganese phosphate**. Same family, three weight classes.

## • WHERE IRON PHOSPHATE WORKS — AND WHERE IT DOESN'T

- **Steel and iron: excellent.** This is iron phosphate's home turf — it's literally built from the iron in the part.
- **Galvanized (zinc-coated) steel and aluminum: limited.** On these metals there's little or no iron to convert, so a true iron phosphate film barely forms. The chemistry mostly just *cleans* those surfaces. Mixed-metal lines (steel + aluminum + galvanized together) are a classic case where shops move to a **zirconium pretreatment** instead (covered later).



### REMEMBER

**Iron phosphate is a steel process.** If your line runs aluminum or galvanized parts and the customer expects a real conversion coating on them, flag it — iron phosphate may be the wrong process for that metal, and a zirconium-based product may be the right call. Confirm with your lead and the chemistry supplier.

## • THE HONEST TRADEOFFS

A good operator knows the weaknesses too:

- **Lighter protection than zinc/manganese phosphate.** Under demanding corrosion specs, iron phosphate alone (even painted) may not pass — those jobs go to zinc phosphate.
- **Best on steel only** (see above).
- **Makes sludge.** As iron dissolves into the bath, it eventually drops out as iron-phosphate sludge that must be filtered/settled and disposed of — a real maintenance and waste item.
- **Phosphate in the wastewater** is a regulated discharge (it feeds algae downstream), so rinses and dumps need attention (covered in Waste & Environment).
- **Being displaced by zirconium “nano” pretreatments** in many modern plants for energy and waste reasons (covered later — and stated honestly, because you may see your shop convert).



### FUN FACT

That shimmering blue-gold color isn't a dye — it's a **thin-film interference** effect, the exact same physics that paints an oil slick or a soap bubble. As the coating thickens by a few millionths of an inch, the color marches from blue toward gold. You're literally reading coating thickness with your eyes.



### REMEMBER

You've got the “why” now: a conversion coating grown by chemistry alone, light and amorphous, read by its blue-to-gold color, whose entire job is to be the grip layer under paint on steel. Keep that picture in your head and every station ahead will make sense.

## CHAPTER 3

# HOW AN IRON PHOSPHATE LINE WORKS

## THE BIG PICTURE — A LINE IS A SYSTEM, NOT A PILE OF SEPARATE STAGES

### A LINE IS A SEQUENCE OF STAGES

An iron phosphate line is not one tank. It's a **sequence of stages** that a part travels through in a strict order. Think of it like a car wash or a dishwasher cycle: each station does one job, and each job prepares the surface for the next. There are two main ways parts move through:

- **Spray washer (very common for paint prep).** Parts hang from an **overhead conveyor** (a “monorail”) or sit in baskets and travel through an enclosed **tunnel**, where banks of nozzles spray each chemistry stage in turn, with rinses between. Fast, continuous, great for high volume. This is the classic iron phosphate setup feeding a paint or powder line.
- **Immersion (dip).** Parts are lowered into a series of tanks in sequence — clean, rinse, phosphate, rinse, seal — then dried. Slower, but gentler and better for complex shapes, blind holes, and recesses where spray can't reach.

**TIP**

Spray and dip use the *same chemistry family* but **different numbers**. Spray contact times are short (often 30–90 seconds) because the constant fresh spray is aggressive and efficient; immersion times run longer (often 1–3 minutes) because the part just sits in still solution. When in doubt about which times apply, check whether you're on a spray line or a dip line and read the matching column on your shop-floor poster.

## THE CLEANER-COATER CONCEPT (IMPORTANT — AND REUSABLE)

Here's a trick unique to the lighter phosphates, and you'll meet it constantly on iron phosphate lines: a **cleaner-coater** is a single product that **cleans and coats in one stage**. It contains both detergents/surfactants (to lift oil and shop soil) *and* the phosphating chemistry (to grow the conversion film). The part comes out clean *and* coated from one tank or one spray zone.

Why does this exist? Because it lets a shop collapse a 5-stage line down to a **3-stage** line:

- **5-stage** (separate clean + separate phosphate): Clean → Rinse → Phosphate → Rinse → Seal
- **3-stage** (cleaner-coater does both): Cleaner-Coater → Rinse → Seal

The cleaner-coater route is cheaper to build and run and uses less floor space — but because it asks one bath to do two jobs, it typically lays down a **lighter** coating (often the ~10–50 mg/ft<sup>2</sup> end) and tolerates less heavily-soiled parts than a dedicated two-step. Heavily oiled or scaled parts, and jobs needing the upper coating-weight range, usually want the full **5-stage** with a dedicated alkaline cleaner up front.

**REMEMBER**

A **cleaner-coater cleans and coats at the same time, in one stage**. That's what makes a 3-stage line possible. It trades a little coating weight and a little dirt-tolerance for lower cost and a smaller footprint. Know which kind of line you're on — it changes how you read every downstream stage.

• THE WHOLE SEQUENCE: 8 STATIONS (TEACHING LAYOUT)

Here’s the complete process as we’ll teach it. Don’t memorize the details — just get a feel for the *flow*. Each station gets its own full chapter.

| #  | STATION                | THE ONE THING IT DOES                                       | TEMP      | TIME               |
|----|------------------------|-------------------------------------------------------------|-----------|--------------------|
| 01 | Inspect & Load         | Confirms the part & hangs it so solution reaches everything | Ambient   | –                  |
| 02 | Alkaline Clean         | Removes oil, grease, shop soil so chemistry can touch steel | 120–160°F | 1–5 min / 30–120 s |
| 03 | Rinse (Post-Clean)     | Washes off cleaner before the phosphate stage               | Ambient   | 30–60 s            |
| 04 | Iron Phosphate         | Grows the conversion film — the paint base                  | Amb–150°F | 1–3 min / 30–90 s  |
| 05 | Rinse (Post-Phosphate) | Washes off spent phosphate; protects the seal               | Ambient   | 30–60 s            |
| 06 | Seal / Final Rinse     | Passivates pores; locks in corrosion resistance under paint | Amb–120°F | 30–60 s            |
| 07 | Dry-Off Oven           | Flashes off <b>all</b> water so paint/powder goes on dry    | 250–400°F | 3–10 min           |
| 08 | Inspect & Hand-Off     | Confirms color/weight; sends a clean, dry part to paint     | Ambient   | –                  |

The basic rhythm: **Inspect** → **Clean** → **Rinse** → **Phosphate** → **Rinse** → **Seal** → **Dry-Off** → **Inspect/Paint**.



TIP

On a 3-stage cleaner-coater line, mentally fold Stations 02 and 04 together (the cleaner-coater does both), drop one rinse, and you’ve got Cleaner-Coater → Rinse → Seal → Dry. Everything else in this manual still applies — the same hazards, the same “why the order matters” logic, the same coating-weight thinking.

• A LINE IS A SYSTEM, NOT A PILE OF SEPARATE TANKS



REMEMBER

**Every stage either protects the next stage or poisons it. There is no neutral step.** A part that leaves the cleaner still carrying a film of cleaner will dilute and contaminate the phosphate. A part that leaves the phosphate still wet will drag phosphate into the seal. What you do (or fail to do) at the cleaner shows up as a paint failure three weeks later in the field. The line is a chain, and a chain is only as strong as its weakest link.



TIP

Think of the phosphate stage as the heart of the line. Everything *before* it (inspect, clean, rinse) exists to deliver a perfectly clean, bare, well-rinsed steel surface for the reaction. Everything *after* it (rinse, seal, dry) exists to protect and finish what the phosphate created. The whole line orbits that one stage — and the paint job downstream lives or dies by it.

## CHAPTER 4

# WHY THE ORDER MATTERS

## WALKING THE LINE — EACH STAGE SETS UP THE NEXT, AND THE SEQUENCE IS LOCKED BY CHEMISTRY

You might think the stage order is just convention — that you could shuffle steps or skip one to save time. You can't. Each stage exists *because of* the stage before and after it. Here's the logic, link by link.

### STATION 02 — CLEAN: GET RID OF THE JUNK FIRST

The phosphate reaction needs to touch *bare steel*. Oil, grease, and shop soil block it. **Phosphate over oil and you've phosphated nothing** — the coating forms patchy or not at all wherever soil remains, and the paint over those spots will fail. There's a free, foolproof check called the **water-break test**: pull a cleaned part and watch the water. If it sheets off in an unbroken film, the surface is clean. If it beads up like rain on a waxed car, oil remains — re-clean.

**TIP**

The simplest, most powerful quality check in the whole shop is the **water-break test**, and it's free. A smooth, continuous, unbroken sheet of water ("zero seconds" to break) means clean. Beading or "breaking" into droplets means oil is still there — *stop and re-clean*. On iron phosphate, a clean surface isn't just nice-to-have; it's the difference between a coating forming and not forming.

**HEADS-UP**

The cleaner is **hot (120–160°F)** and **alkaline (caustic)** — it causes chemical burns and can seriously injure your eyes. Wear full PPE, and know where your shower and eyewash are *before* you need them.

### STATION 03 — RINSE (POST-CLEAN): DON'T CARRY THE CLEANER FORWARD

The cleaner is **alkaline** (high pH). The next stage — iron phosphate — is **acidic** (low pH). Acid and alkaline are chemical opposites: when they meet, they neutralize and cancel each other out. Carry a film of alkaline cleaner into the phosphate and you **raise the phosphate's pH**, weakening it and pushing it out of its narrow working window. The rinse strips the cleaner film off first.

**REMEMBER**

*"Drag cleaner into the phosphate and you blunt the phosphate."* Rinses exist to keep each chemistry from invading the next stage. This is your first firewall on the line.

## STATION 04 — IRON PHOSPHATE: THE MAIN EVENT

Now — and only now, with a clean, bare, well-rinsed steel surface — does the conversion coating actually form. The mildly acidic phosphate solution reacts with the steel and grows the thin, iridescent film that paint will grip (Chapter 1 covered the chemistry). This is the payoff stage.



### HEADS-UP

The phosphate solution is **mildly acidic** (phosphoric-acid based) and contains **oxidizing accelerators** (often nitrite-, chlorate-, or molybdate-based). It's gentler than a plating-line pickle, but it still irritates skin, eyes, and lungs, and the spray throws a fine mist. **The accelerators are oxidizers** — keep them away from incompatible chemicals, and never let spilled accelerator dry on rags or combustibles. Ventilation and PPE are required.

## STATION 05 — RINSE (POST-PHOSPHATE): PROTECT THE SEAL

The part leaves the phosphate stage wet with spent, iron-laden phosphate solution. Drag that into the seal and you contaminate and shorten the life of the seal chemistry, and you can leave salt residues under the paint that cause blistering later. This rinse cleans the freshly coated surface without disturbing the new film.



### REMEMBER

**Phosphate residue under paint is a future blister.** This rinse — ideally ending with clean or **deionized (DI)** water — is the wall between the phosphate stage and the seal. Skimp here and parts can flash-rust or carry salts that paint can't cover for.

## STATION 06 — SEAL: LOCK IN THE CORROSION RESISTANCE

The conversion film is good, but it's slightly porous, and the rinse can leave the surface chemically “open.” The **seal** (also called the **final rinse** or **passivating rinse**) reacts with or coats those pores to dramatically boost under-paint corrosion resistance. Historically this was a dilute **chromic acid** rinse; modern lines use **non-chrome** seals (zirconium-based or organic/polymer). Many seals are “**dried-in-place**” (no rinse after — the part goes straight to dry-off).



### HEADS-UP

If your seal is a **chromic / chromate (hexavalent chromium, Cr(VI))** product, you are handling a **confirmed human carcinogen** with strict OSHA controls (covered in the seal chapter). Many shops have switched to **non-chrome** seals specifically to eliminate this hazard. **Know which seal your shop runs.**

### STATION 07 — DRY-OFF: GET IT BONE DRY

Paint and powder must go on a **completely dry** surface. Any trapped water — in a seam, a blind hole, a weld — will boil out during paint cure and blister or pop the finish. The dry-off oven flashes off every drop before the part reaches paint.



**REMEMBER**

**Wet steel + paint = guaranteed defect.** The dry-off oven isn't optional drying — it's a quality gate. Water hiding in a crevice is one of the most common causes of "mystery" paint blisters downstream.

### STATION 08 — INSPECT & HAND-OFF: SEND IT OUT RIGHT

A final look confirms the coating color/weight is in range and the part is clean and dry, then it goes to paint. What you confirm here is the last chance to catch a problem before it's buried under a finish coat.



**REMEMBER**

The whole eight-station sequence is one continuous logic chain: *clean it so the phosphate can react → rinse so the cleaner doesn't blunt the acid → phosphate to grow the paint base → rinse so residue doesn't poison the seal or blister the paint → seal to lock in corrosion resistance → dry so paint goes on a dry surface → inspect so nothing bad ships.* Every step earns its place. Skip one, and the next step — or the paint line, or the customer — tells on you.



**TIP**

When you understand *why* each stage exists, troubleshooting becomes logical instead of mysterious. Paint peeling? Look upstream at cleaning and phosphate. Blisters under paint? Look at the post-phosphate rinse and the dry-off. Powdery coating? Look at phosphate weight/temperature. The defect almost always points back to the stage that was supposed to prepare for it.

CHAPTER 5

# PPE — YOUR DAILY ARMOR

## THE GEAR THAT STANDS BETWEEN YOU AND CHEMICALS THAT BURN, MIST THAT IRRITATES, OVENS THAT SCORCH

An iron phosphate line works with hot caustic cleaners, mildly acidic phosphate solutions, oxidizing accelerators, possibly chromium seal chemistry, hot ovens, chemical mists, and slick wet floors. None of these will hurt you if you respect them and wear the right gear. Several can hurt you badly if you don't.

 **REMEMBER**  
There is no part, no order, and no deadline worth an injury. Scrap can be reworked. Your eyes cannot.

### THE MASTER PPE SET (WEAR ON THE LINE)

- **Chemical splash goggles** — sealed goggles, not safety glasses. Glasses leave gaps; a splash finds the gaps.
- **Face shield** — over the goggles for any work involving the hot cleaner, the acidic phosphate, the seal, or charging/dumping tanks.
- **Chemical-resistant gloves** — elbow-length, rated for both acids and alkalis (nitrile, neoprene, or PVC per the chemistry). Inspect for pinholes before each use.
- **Chemical-resistant apron** — full-length, chest to below the knee.
- **Chemical-resistant, non-slip boots** — floors around washers and dip tanks are wet and slick.
- **Long sleeves / appropriate clothing** — no exposed skin where splashes or mist are likely.
- **Heat-resistant gloves** for dry-off oven work — different gloves than your chemical gloves.

 **HEADS-UP**  
**Respiratory protection** is required when ventilation is inadequate or when working over/near fuming or misting stages — the alkaline cleaner mist, the acidic phosphate mist, and especially any **chromate seal**. Use the cartridge or supplied-air respirator your shop specifies for that station. If you can smell acid or strongly smell the cleaner, ventilation isn't keeping up — stop and report it.

 **TIP**  
Put your gear on *in order*: boots and apron first, then gloves, then goggles, then face shield last. Take it off in reverse. This keeps you from touching your face with contaminated gloves. Rinse your gloves *before* you take them off.

### STATION-BY-STATION HAZARD SUMMARY

| STATION           | PRIMARY HAZARD                                | WHY IT'S DANGEROUS                                 |
|-------------------|-----------------------------------------------|----------------------------------------------------|
| 01 Inspect/Load   | Sharp/heavy parts; pinch points; conveyor     | Cuts, crush, caught-in                             |
| 02 Alkaline Clean | Hot caustic (120–160°F), mist                 | Chemical burns; eye injury; respiratory irritation |
| 03 Rinse          | Residual cleaner; wet floor                   | Mild irritant; <b>slips</b>                        |
| 04 Iron Phosphate | Mild acid + oxidizing accelerator; spray mist | Burns/irritation; oxidizer hazards; inhalation     |
| 05 Rinse          | Phosphate residue; wet floor                  | Mild irritant; slips                               |
| 06 Seal           | Acidic seal; <b>Cr(VI) if chromate seal</b>   | Burns; Cr(VI) is a <b>carcinogen</b>               |
| 07 Dry-Off Oven   | High heat (250–400°F); hot parts              | Severe burns; heat stress                          |
| 08 Inspect/Paint  | Solvents/paint at hand-off; hot parts         | Flammables; burns                                  |

 **HEADS-UP**  
**Never eat, drink, smoke, vape, or apply cosmetics on the line.** Hand-to-mouth contact is a major route for ingesting process chemicals. Wash thoroughly before breaks and before leaving. Keep food and drink completely out of the work area.

CHAPTER 6

# EMERGENCY & FIRST-RESPONSE

KNOW THESE BEFORE YOUR FIRST SHIFT, NOT DURING YOUR FIRST INCIDENT

 **HEADS-UP**

Memorize the location of the nearest **eyewash, safety shower, fire extinguisher, spill kit, first-aid kit, and emergency exit** *before* you start work. In an emergency you won't have time to go looking.

• **FIRST RESPONSE BY HAZARD**

| SITUATION                                          | WHAT TO DO — IMMEDIATELY                                                                                                                                                                                                                                                                  |
|----------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Chemical on skin (cleaner, phosphate, seal)        | Remove contaminated clothing, flush with water <b>at least 15 minutes</b> , get help. Don't "wipe and continue."                                                                                                                                                                          |
| Chemical in eyes                                   | Go straight to the eyewash, hold eyelids open, flush <b>at least 15 minutes</b> , get medical help. Eyes don't get second chances.                                                                                                                                                        |
| Inhaled mist/fumes (dizzy, throat/lung irritation) | Get to fresh air immediately; report it. The ventilation may be failing.                                                                                                                                                                                                                  |
| Burn from the dry-off oven or a hot part           | Cool with water; do not peel clothing stuck to a burn; get medical help for anything beyond minor.                                                                                                                                                                                        |
| Spill                                              | Only clean up small spills you're trained for, in PPE. Acidic phosphate and alkaline cleaner are <i>opposites</i> — never just mix them to "cancel out"; follow your shop's spill procedure. <b>Oxidizing accelerator spills</b> need special care — keep them off combustibles and rags. |
| Any chromate (Cr(VI)) seal exposure                | Treat as serious; flush, report, and follow the specific SDS and your shop's Cr(VI) exposure procedure.                                                                                                                                                                                   |

 **REMEMBER**

When in doubt, **flush, then ask**. Water and time fix most chemical contact if you act fast. Reporting every incident — even "minor" ones — is how the shop fixes the cause before it hurts the next person.

 **HEADS-UP — LOCKOUT/TAGOUT (LOTO)**

There's no current through the parts, but the washer's pumps, heaters, conveyor, and oven are powered by serious electrical equipment. **LOTO is mandatory before any maintenance** — physically lock the power off and tag it so no one can switch it back on while you're working. "No current at the part" does not mean "no current in the building."

 **FUN FACT**

The 15-minute flush rule isn't arbitrary — it's the time emergency-response standards found necessary to dilute and carry away most corrosive chemistry before it does deep tissue damage. That's why eyewash and shower stations are required to deliver a steady flow for a full 15 minutes hands-free.

## CHAPTER 7

# THE SAFETY PHILOSOPHY OF AN IRON PHOSPHATE LINE

## DIFFERENT HAZARDS THAN A PLATING LINE — NOT FEWER

People sometimes assume that because there's no high-amperage rectifier at the parts, a pretreatment line is “the safe one.” That's a dangerous assumption. The hazards just *moved*:

- **Heat is everywhere.** Hot cleaners, sometimes hot phosphate, and a **250–400°F dry-off oven** are the biggest burn sources on the line.
- **Mist is the sneaky one.** Spray washers atomize every chemistry stage into a fine mist. You're not splashing it on — you're potentially *breathing* it. Ventilation is a primary control, not an afterthought.
- **The accelerators are oxidizers.** They make the chemistry work, but oxidizers demand respect around fuels, rags, and incompatible chemicals.
- **The seal may be a carcinogen.** Chromate (Cr(VI)) seals carry the most serious chronic-health hazard on the entire line. Non-chrome seals exist largely to remove that risk.
- **The washer tunnel is a confined space.** Climbing inside to clear nozzles or service the tunnel can be **confined-space entry** — with its own permit, ventilation, atmosphere testing, and LOTO requirements.



### HEADS-UP — CONFINED SPACE: THE WASHER TUNNEL

A spray washer tunnel can meet the definition of a **permit-required confined space**: limited entry/exit, not designed for continuous occupancy, and possible hazardous atmosphere (chemical mist, low oxygen, hot/humid). **Never crawl into a washer to clear nozzles or do maintenance without your shop's confined-space procedure**: LOTO of pumps/heaters/conveyor, atmosphere testing, ventilation, an attendant outside, and a permit. People have been seriously hurt entering “just for a second.”



### REMEMBER

The safety philosophy of this line is simple: **respect the heat, control the mist, know your seal chemistry, and never enter the tunnel casually**. A pretreatment line isn't “the safe line” — it's a *different* line with its own hazards that deserve the same discipline you'd give a tank full of acid.



### REMEMBER

You've now got the whole foundation: what a conversion coating is, what makes iron phosphate special, how the eight-station line works as one connected system, why the order is locked by chemistry, what to wear, what to do in an emergency, and the safety mindset that ties it together. Everything in the station chapters builds on these fundamentals. Turn the page knowing the *why* — and the *how* will make a lot more sense.

# INSPECT & LOAD

“GARBAGE IN, GARBAGE OUT — THE LINE CAN ONLY FINISH WHAT YOU HANG ON IT.”

Before any chemistry touches a part, someone has to confirm it's the right part, in the right condition, and load it so the line can actually reach every surface. On a conveyor line this is the **load station** (also called load/unload, “load horse,” or hang-on); on a dip line it's racking or basketing. It feels like the un-glamorous step. It isn't: a part hung wrong gets cleaned wrong, coated wrong, and painted wrong.

## • WHAT YOU'RE DOING & WHY IT MATTERS

- **Confirm the job.** Right part number, right quantity, right spec. Check the traveler for the **target coating weight/color**, the **substrate** (is it actually steel?), and any special instructions.
- **Inspect the incoming condition.** Heavy rust, mill scale, weld spatter, heavy grease, or dried-on coolant may need pre-handling — the standard cleaner won't fix heavy scale (that's a pickling/blasting job, not iron phosphate).
- **Load for full contact and full drainage.** Every surface must see solution, and every cavity must **drain** — trapped solution in a cup, channel, or blind hole becomes a contamination pocket and a paint defect.

## • OPERATOR ESSENTIALS

| ITEM            | GUIDANCE                                                                               |
|-----------------|----------------------------------------------------------------------------------------|
| Substrate check | Steel/iron = good. Galvanized/aluminum = flag it (iron phosphate may be wrong process) |
| Orientation     | Hang so liquids <b>drain freely</b> — no cups, pockets, or trapped air                 |
| Spacing         | Parts not touching/nesting — solution and spray must reach all faces                   |
| Load weight     | Within hanger/conveyor rating; balanced                                                |
| Pre-clean flags | Heavy scale, rust, weld spatter, thick grease → route for special handling             |



### HEADS-UP

**Mechanical hazards rule this station.** Sharp edges, heavy parts, pinch points at hangers, and a **moving conveyor**. Wear cut-resistant gloves for handling, watch your hands around hooks and the rail, and never reach into a moving conveyor. Lift with your legs; get help for heavy loads.

• **STEP-BY-STEP PROCEDURE**

- 1. **Read the traveler.** Confirm part, quantity, substrate, and target coating weight/color.
- 2. **Inspect each part** for heavy soil, scale, rust, or damage. Set aside anything needing special handling.
- 3. **Hang or load** so every surface is exposed and every cavity drains. Avoid nesting and trapped-air pockets.
- 4. **Confirm spacing and balance** so parts don't bang together or fall on the line.
- 5. **Note anything unusual** (mixed substrates, oily parts) to your lead before the load enters the cleaner.

• **TOP MISTAKES NEW OPERATORS MAKE**

- 1. **Hanging parts that trap solution** in cups/channels — guaranteed drips, spots, and blisters later.
- 2. **Nesting parts** so faces touch — those faces come out uncleaned and uncoated.
- 3. **Loading the wrong substrate** without flagging it — aluminum/galvanized won't coat like steel.
- 4. **Overloading** the hanger/conveyor — parts fall or block spray.
- 5. **Ignoring heavy scale/rust** and expecting the cleaner to handle it — it can't.

• **TEACH-BACK CHECKLIST**

- ☐ Explain why drainage orientation matters (trapped solution = contamination/blisters).
- ☐ State which substrate iron phosphate is designed for (steel) and what to do with aluminum/galvanized (flag it).
- ☐ Identify two mechanical hazards at this station (sharp edges, pinch points, moving conveyor).
- ☐ Describe what incoming conditions need special handling (heavy scale, rust, weld spatter, thick grease).

**SIGN-OFF — STATION 01**

Trainee: \_\_\_\_\_ Trainer: \_\_\_\_\_ Date: \_\_\_\_\_

*"I confirm I verified the job and substrate, inspected incoming condition, and loaded parts for full contact and free drainage, with safe handling and PPE."*

STATION 02 OF 08

# ALKALINE CLEAN / DEGREASE

“PHOSPHATE OVER DIRT AND YOU’VE PHOSPHATED DIRT.”

This is where oil, grease, fingerprints, and shop soil come off so the steel is bare and reactive. On a 5-stage line it’s a **dedicated alkaline cleaner**; on a 3-stage line the **cleaner-coater** does this job *and* the phosphating in one stage (see Station 04 for that variant). Either way, **clean steel is non-negotiable** — the phosphate reaction simply won’t happen on an oily surface.

## • WHAT YOU’RE DOING & WHY IT MATTERS

The cleaner is an **alkaline (high-pH) detergent solution**, usually warm, that does three things: **emulsifies and lifts oil/grease**, **suspends particulate soil** so it rinses away, and **wets the surface** so nothing redeposits. The result is a uniformly clean, water-loving surface that passes the **water-break test**.

 **TECHNICAL STUFF**

Iron phosphate alkaline cleaners are often **milder** (pH ~9–12) than a heavy electroplating soak cleaner, because the goal is degreasing without aggressively etching, and because some lines run mixed metals that strong caustic would attack. Builders/buffers, surfactants, and sometimes mild chelators do the work. Exact concentration and pH come from the **supplier TDS** — never guess.

## • OPERATOR ESSENTIALS

| PARAMETER                  | RANGE                        | NOTES                                                 |
|----------------------------|------------------------------|-------------------------------------------------------|
| Temperature                | 120–160°F (49–71°C)          | Warmth speeds degreasing; <b>don’t exceed TDS max</b> |
| Time                       | 1–5 min dip / 30–120 s spray | Heavier oil = longer end of range                     |
| Concentration              | Per supplier TDS (titrate)   | Too weak won’t cut oil; too strong wastes chemical    |
| pH                         | ~9–12 (per product)          | Milder than a plating soak clean                      |
| Agitation / spray pressure | Per equipment                | Movement keeps fresh cleaner on the surface           |
| Oil skimming / control     | Keep surface oil skimmed     | Floating oil redeposits on parts                      |

 **HEADS-UP — HOT ALKALINE SOLUTION**

This tank/zone is **hot and caustic**. Splashes cause **chemical burns on contact**, and the steam/mist irritates airways. **PPE every time:** chemical splash goggles, face shield (when charging/dumping), elbow-length chemical gloves, apron, chemical-resistant boots. First aid: skin → flush 15 min; eyes → eyewash 15 min and get help. Never lean over the tank without eye protection.

• STEP-BY-STEP PROCEDURE

- 1. **Check the bath before you start.** Confirm temperature is in range and concentration is in spec (last titration log). Make sure surface oil is skimmed.
- 2. **Inspect the load.** Heavy oil → plan for the long end of the time range. Light oil → short end.
- 3. **Clean fully** — immerse or run through the spray zone for the full time. Confirm agitation/spray is working and reaching all faces.
- 4. **Withdraw and (if dip) let it drain** back into the tank.
- 5. **Run the water-break test** on a sample part: continuous sheet = pass; beading = fail, re-clean.
- 6. **Move promptly** to the rinse so the clean surface doesn't sit and flash-rust.



**TIP — THE WATER-BREAK TEST (FREE, FAST, NEVER LIES)**

A truly clean steel surface loves water, so it **sheets off in one continuous, unbroken film**. A dirty surface repels water, so it **beads up** into droplets. **Continuous sheet = PASS** (send it on). **Beading = FAIL** (re-clean). On iron phosphate this isn't a nicety — it's the difference between a coating forming and not forming.

• TOP MISTAKES NEW OPERATORS MAKE

- 1. **Skipping the water-break test** because the part “looks fine.” The oil that ruins phosphating is invisible.
- 2. **Running the bath too cold or too weak** — it can't cut oil no matter how long you soak. Check temp and titration first.
- 3. **Letting surface oil build up** with no skimming — it redeposits on the way out.
- 4. **Nesting parts** so cleaner can't reach every face.
- 5. **Letting clean parts sit** before rinsing — flash rust and re-soil.

• QUICK TROUBLESHOOTING

| SYMPTOM                     | PROBABLE CAUSE                     | WHAT TO DO                           |
|-----------------------------|------------------------------------|--------------------------------------|
| Water-break fails           | Low concentration or temperature   | Titrate; bring temp to range         |
| Oily film after cleaning    | Floating oil redepositing          | Skim; refresh/dump if heavily loaded |
| Patchy phosphate downstream | Uneven cleaning, trapped air       | Reposition load; check spray pattern |
| White residue on parts      | Hard-water salts / cleaner residue | Check rinse water; improve rinse     |

**SIGN-OFF — STATION 02**

Trainee: \_\_\_\_\_ Trainer: \_\_\_\_\_ Conc./Temp: \_\_\_\_\_

*“I confirm parts passed the water-break test, the cleaner was in concentration and temperature range, and I wore required PPE.”*

STATION 03 OF 08

# RINSE (POST-CLEAN)

“A RINSE ISN’T A REST STOP. IT’S A CHECKPOINT.”

You just got the steel clean — but it’s coated in a film of alkaline cleaner. This station’s whole job is to **wash that cleaner film off** before the part hits the acidic phosphate stage. It looks like a “nothing” step. It isn’t: a weak rinse here quietly raises the phosphate bath’s pH and pushes it out of its narrow window.

## • WHAT YOU’RE DOING & WHY IT MATTERS

The cleaner is **alkaline**; the phosphate is **acidic**. Carry cleaner forward and it **neutralizes acid and raises the phosphate’s pH**, weakening the coating reaction. The invisible carry-over of solution clinging to a part is **drag-out** (leaving a tank) / **drag-in** (arriving at the next). The rinse knocks drag-out down to almost nothing by dilution: clean water surrounds the part, the cleaner film dissolves into the rinse, and constant flow flushes it away.

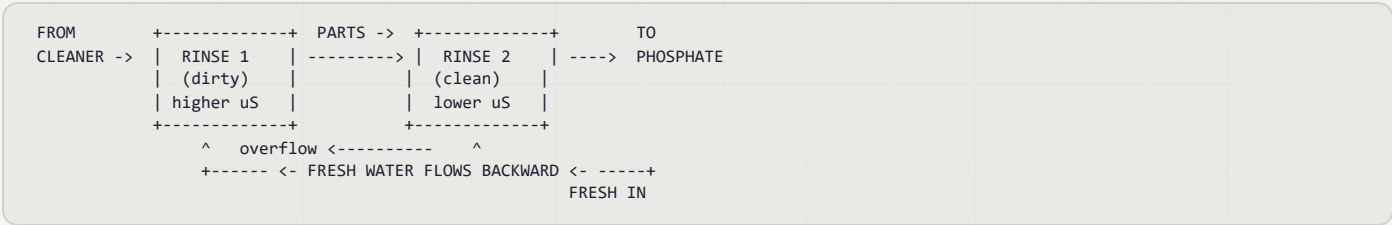


### TIP — HOW WE MEASURE A CLEAN RINSE: CONDUCTIVITY

You can’t see dissolved cleaner, so we measure **conductivity** — how well the water conducts electricity. Pure water barely conducts; dissolved chemicals make it conduct more. So **higher conductivity = more cleaner left in the rinse = dirtier rinse**. It’s measured in **microsiemens per centimeter (µS/cm)** — just think “the contamination number.” Lower is cleaner.

## • THE SMART WAY TO RINSE: COUNTER-FLOW

You *could* blast fresh water through one tank and dump it — it works, but wastes enormous water. The professional approach is a **counter-flow (counter-current) rinse**: parts move forward, water flows backward. Fresh water enters the *last* (cleanest) stage and overflows backward into the dirtier one, so the cleanest water the part ever sees is the last thing to touch it before the phosphate.



### HEADS-UP

Rinse water carries dragged-in cleaner and starts out mildly caustic. Wear **goggles, chemical gloves, and non-slip footwear** — the floor around rinses is always wet, and **slips are the #1 injury here**.

• OPERATOR ESSENTIALS

| PARAMETER    | RANGE                                 | NOTES                           |
|--------------|---------------------------------------|---------------------------------|
| Temperature  | Ambient (60–85°F)                     | No heating required             |
| Time         | 30–60 s with agitation/spray          | Longer for blind holes/tubing   |
| Water source | Municipal or DI                       | DI rinses cleaner, fewer spots  |
| Conductivity | Keep low (e.g. < 500 µS/cm)           | Monitor daily; rising = drag-in |
| Flow         | Continuous overflow / counter-current | Counter-flow saves water        |
| pH           | Near neutral (7–9)                    | High pH = cleaner drag-in       |

• STEP-BY-STEP PROCEDURE

1. **Check conductivity at shift start.** Rising/high → tell your lead; the rinse needs more flow or a refresh.
2. **Confirm water is flowing** (overflow or counter-current active).
3. **Transfer promptly** from the cleaner — dried cleaner is harder to rinse off.
4. **Rinse fully, 30–60 s**, with agitation/spray; flush blind holes and tubing longer.
5. **Drain** briefly to carry less forward.
6. **Move promptly** to the phosphate stage — don't let the part flash-rust.

• TOP MISTAKES NEW OPERATORS MAKE

1. Treating the rinse as “quick” — a two-second dunk rinses nothing.
2. Ignoring conductivity — a high number means you're blunting the phosphate right now.
3. Letting the part drip-dry between stages — dried cleaner is stubborn.
4. Not flushing blind holes/tubes — concentrated cleaner hides and rides forward.

• TEACH-BACK CHECKLIST

- ☐ Explain why carrying cleaner into the phosphate causes problems (raises pH, weakens coating).
- ☐ Define drag-out / drag-in.
- ☐ Explain what conductivity tells you and which direction is “clean” (lower = cleaner).
- ☐ State the rinse time and why tubing needs longer.

SIGN-OFF — STATION 03

Trainee: \_\_\_\_\_ Trainer: \_\_\_\_\_ Conductivity: \_\_\_\_\_ µS/cm

*“I confirm rinse conductivity was within limit, water flow was running, and parts were rinsed the full time with PPE worn.”*

STATION 04 OF 08 • THE MAIN EVENT

# IRON PHOSPHATE (OR CLEANER-COATER)

“THIS IS THE STATION WHERE BARE STEEL BECOMES A PAINT BASE.”

Now — with a clean, bare, well-rinsed steel surface — the conversion coating actually forms. The mildly acidic phosphate solution reacts with the steel and grows the thin, iridescent iron-phosphate/iron-oxide film that paint and powder grip. Everything before this stage existed to deliver a perfect surface here; everything after exists to protect what happens here.



## REMEMBER — THE TWO WAYS THIS STAGE SHOWS UP

On a **5-stage line** this is a dedicated **iron phosphate** stage (the part was already cleaned at Station 02). On a **3-stage line** this same stage is a **cleaner-coater** that *cleans and coats together* — in that case Station 02 doesn't exist as a separate step. Same chemistry family, same hazards, same coating-weight thinking; just combined.

## • WHAT YOU'RE DOING & WHY IT MATTERS

The bath is dilute **phosphoric acid** plus **alkali-metal phosphate salts**, buffered to a narrow **pH ~4.5–5.5**, with an **accelerator** (an oxidizer) to drive the reaction and a **wetting agent**. The acid lightly pickles the steel; the local pH rise precipitates the amorphous film (Chapter 1). The accelerator keeps it fast and uniform and helps build the iridescent blue-to-gold color that tells you the weight is right.



## TECHNICAL STUFF — BATH CONTROL

Operators control the phosphate bath with simple titrations: **Free acid (points)** — the amount of *unreacted* acid (too high = coating won't form / too aggressive; too low = sluggish, powdery coatings). **Total acid (points)** — total acid content (free + combined); the **ratio of total acid to free acid** is a key health indicator. **Accelerator level** — measured by its own test (a drop or strip test for the oxidizer); low accelerator = slow, thin, or no coating. Run these per your supplier's TDS — the “points” are product-specific.



## HEADS-UP — MILD ACID + OXIDIZING ACCELERATOR + MIST

The solution is acidic (irritant to skin/eyes/lungs) and the spray makes a fine mist you must not breathe — **local exhaust ventilation is required**. The **accelerator is an oxidizer**: keep it away from fuels, solvents, and rags; never let spills dry on combustibles; store and handle per its SDS. Some accelerators (nitrite-based) have their own handling cautions. PPE: goggles + face shield, chemical gloves, apron, boots, respirator if ventilation is inadequate.

• OPERATOR ESSENTIALS

| PARAMETER              | RANGE                         | NOTES                                      |
|------------------------|-------------------------------|--------------------------------------------|
| pH                     | 4.5–5.5                       | Narrow window; the heart of control        |
| Temperature            | Amb–150°F (low-temp 70–120°F) | Per product; low-temp versions save energy |
| Time                   | 1–3 min dip / 30–90 s spray   | Spray is faster/more aggressive            |
| Target coating weight  | ~30–90 mg/ft² (0.3–1.0 g/m²)  | Read as blue→gold color; confirm to spec   |
| Free acid / total acid | Per TDS (“points”)            | Titrate on schedule                        |
| Accelerator            | Per TDS                       | Test and replenish; low = no coating       |
| Sludge control         | Filter/settle regularly       | Iron-phosphate sludge builds up            |

• READING THE COATING BY COLOR (YOUR FAST FIELD CHECK)

| COLOR                    | MEANING                        | ACTION                                                      |
|--------------------------|--------------------------------|-------------------------------------------------------------|
| Gray / no color          | Too light — not enough coating | Check temp, time, free acid, accelerator                    |
| Blue / blue-iridescent   | Light, often acceptable        | Verify to spec                                              |
| Gold / golden iridescent | Medium — strong paint base     | Usually the target band                                     |
| Tan / brown / powdery    | Too heavy / over-reacted       | Lower temp/time; check free acid; <b>don’t ship powdery</b> |



FUN FACT

That shimmering blue-gold color isn’t a dye — it’s a **thin-film interference** effect, the exact same physics that paints an oil slick or a soap bubble. As the coating thickens by a few millionths of an inch, the color marches from blue toward gold. You’re literally reading coating thickness with your eyes.



REMEMBER

Two failure directions: **too light** (bare gray = no grip) and **too heavy** (tan/powdery = a loose layer paint peels off with). Heavier is *not* better. The target is the iridescent blue-to-gold middle — always to your job spec.

• STEP-BY-STEP PROCEDURE

- 1. **Confirm bath health:** pH 4.5–5.5, temperature in range, last titration (free/total acid) and accelerator in spec, sludge filtered.
- 2. **Confirm ventilation** is running and PPE (including face shield) is on.
- 3. **Confirm the part arrived clean and well-rinsed** (Station 03). A poor clean here = no coating.
- 4. **Process for the correct time** (dip 1–3 min / spray 30–90 s) for this part and bath.
- 5. **Check the color** on a sample: even blue-to-gold iridescence, no bare gray, no tan/powder.
- 6. **Move promptly to rinse** — don't let phosphate dry on the part.
- 7. **Log** bath checks and any color/weight concerns to your lead.

• TOP MISTAKES & QUICK TROUBLESHOOTING

| SYMPTOM                | PROBABLE CAUSE                                           | WHAT TO DO                                                |
|------------------------|----------------------------------------------------------|-----------------------------------------------------------|
| No coating / bare gray | Low accelerator, high free acid, low temp, or dirty part | Test accelerator/free acid; verify temp; check Station 02 |
| Powdery / tan / brown  | Coating too heavy; free acid off; temp too high          | Lower temp/time; adjust free acid per TDS                 |
| Uneven / patchy color  | Poor cleaning or poor spray coverage                     | Re-clean upstream; check nozzles/load orientation         |
| Coarse / rough film    | High sludge / contamination                              | Filter or refresh bath                                    |

• TEACH-BACK CHECKLIST

- ☐ Explain how iron phosphate forms with **no electric current**.
- ☐ State the target coating-weight range (~30–90 mg/ft²) and what blue / gold / tan mean.
- ☐ State the pH window (4.5–5.5) and name the three control checks (free acid, total acid, accelerator).
- ☐ Explain why “heavier is not better” (powdery = poor adhesion).
- ☐ Name the accelerator’s hazard class (oxidizer) and the ventilation requirement.
- ☐ Explain the cleaner-coater variant (cleans + coats in one stage → 3-stage line).

SIGN-OFF — STATION 04

Trainee: \_\_\_\_\_ Trainer: \_\_\_\_\_ pH / Color: \_\_\_\_\_

*“I confirm the phosphate bath was in pH/temp/acid/accelerator spec, coating color was an acceptable blue-to-gold (no powder), ventilation ran, and I wore required PPE.”*

STATION 05 OF 08

# RINSE (POST-PHOSPHATE)

“THE WALL BETWEEN THE COATING AND EVERYTHING THAT FOLLOWS IT.”

The freshly coated part leaves the phosphate stage wet with spent, iron- and acid-laden solution. This rinse removes that residue so it can't contaminate the seal, leave salts under the paint, or cause flash rust. Ideally the *last* rinse before the seal uses **clean or deionized (DI) water**, because what's left on the surface here ends up *under the paint forever*.

## • WHAT YOU'RE DOING & WHY IT MATTERS

Two jobs: **stop the phosphating reaction** by washing away the acidic solution, and **deliver a clean, residue-free surface** to the seal. Leftover phosphate salts are hygroscopic (they attract moisture) — trapped under paint, they pull in water and cause **blisters** down the road.

## • OPERATOR ESSENTIALS

| PARAMETER    | RANGE                            | NOTES                      |
|--------------|----------------------------------|----------------------------|
| Temperature  | Ambient                          | No heating required        |
| Time         | 30–60 s with agitation/spray     | Longer for cavities        |
| Water        | Final rinse: <b>DI preferred</b> | Low residue = better paint |
| Conductivity | Low (target per shop)            | Rising = phosphate drag-in |
| Flow         | Continuous / counter-current     | Same logic as Station 03   |

**HEADS-UP**

Mild phosphate residue and wet floors. Goggles, gloves, non-slip footwear. **Don't let rinsed parts sit** — bare freshly-coated steel can flash-rust before it reaches the seal.

## • STEP-BY-STEP PROCEDURE

1. **Confirm flow and conductivity** at shift start.
2. **Rinse fully, 30–60 s**, flushing cavities.
3. **Use the cleanest/DI water last** before the seal.
4. **Drain briefly**, then move promptly to the seal — no sitting, no flash rust.

## • TOP MISTAKES & TEACH-BACK

**TOP MISTAKES TO AVOID**

- Skimping on rinse → phosphate salts under paint → blisters.
- Using hard/dirty water for the *final* rinse → spots and residue.
- Letting parts sit and flash-rust between rinse and seal.

**TEACH-BACK — CAN YOU ANSWER?**

- What this rinse removes & why residue blisters paint?
- Why DI water is preferred for the final rinse?
- The flash-rust risk of letting parts sit?

**SIGN-OFF — STATION 05**

Trainee: \_\_\_\_\_ Trainer: \_\_\_\_\_ Conductivity: \_\_\_\_\_ µS/cm

*“I confirm the post-phosphate rinse was flowing and in range, the final rinse was clean/DI, and parts moved promptly to the seal with PPE worn.”*

STATION 06 OF 08

# SEAL / FINAL RINSE

“THE LAST CHEMISTRY — THE DIFFERENCE BETWEEN ‘PAINTED’ AND ‘PAINTED AND PROTECTED.’”

The conversion film is good, but slightly porous and chemically “open” after rinsing. The **seal** (a.k.a. final rinse, passivating rinse, or after-rinse) reacts with or coats those pores to dramatically boost **under-paint corrosion resistance**. It’s a small step with a big payoff on salt-spray performance. Many seals are “**dried-in-place**” — no rinse after; the sealed part goes straight to the dry-off oven.

## • TWO FAMILIES OF SEAL — KNOW WHICH YOU RUN

- **Chromic / chromate seal (older, Cr(VI))**. A dilute hexavalent-chromium rinse — historically excellent at boosting corrosion resistance. **But Cr(VI) is a confirmed human carcinogen** with strict OSHA controls. Increasingly phased out.
- **Non-chrome seal (modern, preferred)**. **Zirconium-** or **titanium-based**, or **organic/polymer** reactive rinses. Comparable performance for most jobs, far lower hazard, easier waste handling. Most new and converted lines use these.



### HEADS-UP — HEXAVALENT CHROMIUM (CR(VI)) — READ TWICE

If your seal is a chromic/chromate product, you are handling a **confirmed human carcinogen**. OSHA regulates it under **29 CFR 1910.1026** with a very low permissible exposure limit, required exposure monitoring, respiratory protection, medical surveillance, and strict housekeeping. Mist control is critical. **Know which seal your shop uses**. If it’s chromate, follow the full Cr(VI) program — this goes far beyond ordinary line PPE. Many shops switched to **non-chrome (zirconium) seals specifically to eliminate this hazard**.

## • OPERATOR ESSENTIALS

| PARAMETER       | RANGE                                                | NOTES                        |
|-----------------|------------------------------------------------------|------------------------------|
| Type            | Non-chrome (Zr/Ti or polymer) preferred; or chromate | Per shop/spec                |
| pH              | Per product (often mildly acidic)                    | Follow TDS                   |
| Temperature     | Ambient–120°F                                        | Per product                  |
| Time            | 30–60 s                                              | Per product                  |
| Concentration   | Per TDS (titrate/test)                               | Drives corrosion performance |
| Dried-in-place? | Many seals: <b>no rinse after</b>                    | Goes straight to dry-off     |

## • STEP-BY-STEP PROCEDURE

1. **Confirm which seal chemistry** you're running and the matching PPE/ventilation (full Cr(VI) program if chromate).
2. **Check concentration/pH/temperature** per TDS.
3. **Apply the seal** for the correct time.
4. **Follow the correct rinse rule:** dried-in-place seals get **no** rinse; rinse-type seals get the specified rinse.
5. **Move to dry-off.**

## • TOP MISTAKES NEW OPERATORS MAKE

1. **Not knowing the seal chemistry** — and under-protecting against Cr(VI).
2. **Rinsing a dried-in-place seal off** (or failing to rinse a rinse-type) — both ruin performance.
3. **Letting seal concentration drift** — quietly tanks your salt-spray numbers.

## • TEACH-BACK CHECKLIST

- ☐ Explain what the seal does for under-paint corrosion resistance.
- ☐ Name the two seal families and which is preferred and why (non-chrome; avoids Cr(VI) carcinogen).
- ☐ State the special hazard and controls for a chromate (Cr(VI)) seal.
- ☐ Explain "dried-in-place" and the no-rinse rule.



### TIP

The seal is invisible work — you can't see corrosion protection going on. That's exactly why concentration control and knowing your chemistry matter so much: the payoff shows up weeks later in salt-spray and field performance, long after the part has shipped. Treat the seal log as seriously as the phosphate log.

### SIGN-OFF — STATION 06

Trainee: \_\_\_\_\_ Trainer: \_\_\_\_\_ Seal type: \_\_\_\_\_

*"I confirm I identified the seal chemistry, applied it in spec, followed the correct rinse/no-rinse rule, and wore the required PPE (including the full Cr(VI) program if chromate)."*

STATION 07 OF 08

# DRY-OFF OVEN

“WET STEEL AND PAINT DO NOT MIX — EVER.”

Paint and powder must go on a **completely dry** surface. The dry-off oven flashes off every drop of water — including water hiding in seams, blind holes, and welds — before the part reaches paint. It’s the line’s last quality gate before the finish.

## • WHAT YOU’RE DOING & WHY IT MATTERS

The oven heats the part above water’s boiling point long enough to drive off **all** surface and trapped moisture. Skip it, rush it, or under-temp it, and trapped water boils out during paint cure — causing **blisters, pops, craters, and pinholes** in the finish. (Note: this is the *dry-off* oven, separate from the later *cure* oven that bakes powder/paint.)

## • OPERATOR ESSENTIALS

| PARAMETER              | RANGE                        | NOTES                                   |
|------------------------|------------------------------|-----------------------------------------|
| Part metal temperature | 250–400°F (121–204°C)        | Enough to flash off all water           |
| Time                   | 3–10 min                     | Heavier parts need longer to reach temp |
| Goal                   | Bone dry, including cavities | Then to paint while still warm/clean    |

**HEADS-UP — HIGH HEAT: BURN AND HEAT-STRESS HAZARD**

Oven surfaces and exiting parts are **hot enough to cause severe burns**. Use **heat-resistant gloves** (not your chemical gloves), watch for hot parts and racks, and respect heat-stress in the area. Gas-fired ovens add **combustion/ventilation** concerns — never bypass burner safety interlocks. LOTO before any oven maintenance.

**REMEMBER**

A part can look dry on the outside and still hold water in a crevice. The oven’s job is the water you *can’t* see. Most “mystery” paint blisters trace back to incomplete dry-off.

## • STEP-BY-STEP PROCEDURE

1. **Confirm oven temperature** is at setpoint and airflow/exhaust is running.
2. **Confirm dwell time** is adequate for the part mass.
3. **Verify parts exit dry**, including cavities (no steaming, no water droplets).
4. **Move to paint** while clean and dry — don’t let dried parts collect dust or re-condense.

## • TOP MISTAKES & TEACH-BACK

**TOP MISTAKES TO AVOID**

- Under-temp / under-time → trapped water → paint blisters.
- Touching hot parts without heat gloves.
- Letting dry parts sit and pick up dust or humidity before paint.

**TEACH-BACK — CAN YOU ANSWER?**

- Why parts must be fully dry before paint (trapped water blisters)?
- The dry-off temperature/time targets?
- Dry-off oven vs. cure oven — the difference?
- The heat hazards and correct gloves?

**SIGN-OFF — STATION 07**

Trainee: \_\_\_\_\_ Trainer: \_\_\_\_\_ Oven temp: \_\_\_\_\_

“I confirm parts exited the dry-off fully dry (including cavities), oven temp/time were in range, and I handled hot parts with heat-resistant PPE.”

STATION 08 OF 08 • THE FINISH LINE

# INSPECT & HAND-OFF TO PAINT

“THE LAST LOOK BEFORE IT’S BURIED UNDER A FINISH COAT.”

A final inspection confirms the coating is right and the part is clean, dry, and undamaged, then it goes to paint or powder. This is the last chance to catch a problem before it’s invisible under the finish — and the cheapest place to catch it.

## WHAT YOU’RE DOING & WHY IT MATTERS

- **Confirm coating color/weight** — even blue-to-gold iridescence (no bare gray, no powdery tan).
- **Confirm dryness and cleanliness** — no water, no dust, no fingerprints (handle with clean gloves; skin oils ruin adhesion).
- **Confirm timing** — sealed, dried steel shouldn’t sit indefinitely before paint; follow your shop’s max hold time.



### TIP

Once you trust your line, color *is* your inspection. A consistent blue-to-gold sheen across every part is your green light. The moment color drifts gray (too light) or tan/powdery (too heavy), stop and look upstream at Station 04 before you send more parts to paint.

## OPERATOR ESSENTIALS

| CHECK                | PASS LOOKS LIKE                       |
|----------------------|---------------------------------------|
| Coating color/weight | Even blue-to-gold; matches spec       |
| Dryness              | No water anywhere, including cavities |
| Cleanliness          | No dust, oil, or fingerprints         |
| Damage               | No handling marks, scratches, rust    |
| Hold time            | Within shop limit before painting     |



### HEADS-UP

Parts may still be warm; the paint area adds **flammable solvents/coatings**. Handle with clean gloves (skin oils cause paint failures), and observe paint-area fire/ventilation rules.

## STEP-BY-STEP & TEACH-BACK

### STEP-BY-STEP

- Inspect color/weight against the spec.
- Confirm dry and clean, handling only with clean gloves.
- Reject/route back anything bare-gray, powdery, wet, or contaminated.
- Hand off to paint within the hold-time window.

### TOP MISTAKES TO AVOID

- Passing powdery or bare-gray parts — guaranteed paint failure.
- Bare-handing parts — fingerprints become paint defects.
- Letting parts sit too long — flash rust / dust pickup.

### SIGN-OFF — STATION 08

Trainee: \_\_\_\_\_ Trainer: \_\_\_\_\_ Date: \_\_\_\_\_

*“I confirm I inspected coating color/weight, verified parts were clean and dry, handled with clean gloves, and only passed conforming parts to paint.”*

PART THREE — ACROSS THE LINE  
— THE NUMBER THAT MATTERS MOST

# COATING WEIGHT : CONTROL & MEASUREMENT

TOO LIGHT AND PAINT WON'T GRIP; TOO HEAVY AND IT PEELS OFF WITH THE COATING

Coating weight is *the* number for iron phosphate — too light and paint won't grip, too heavy and the coating is powdery and paint peels with it. Target: ~30–90 mg/ft<sup>2</sup> (0.3–1.0 g/m<sup>2</sup>), confirmed to your spec.

## • TWO WAYS OPERATORS JUDGE IT

### 1. BY COLOR (FAST, FREE, EVERY SHIFT)

Gray = too light; blue→gold = in range; tan/powdery = too heavy. This is your continuous, on-the-line check — the one you use every load.

### 2. BY GRAVIMETRIC MEASUREMENT (THE LAB NUMBER)

The reference method:

1. Cut/select a test panel of known surface **area**, processed alongside the parts.
2. **Weigh it** (precise balance).
3. **Strip the coating** in the supplier-specified stripping solution (handle per SDS — strippers are hazardous).
4. **Re-weigh** the stripped panel.
5. **Coating weight** = (weight before – weight after) ÷ area, expressed in mg/ft<sup>2</sup> or g/m<sup>2</sup>.



#### TECHNICAL STUFF

Coating weight is what specs actually call out, and the governing document (a customer spec or a standard such as the federal pretreatment spec **TT-C-490**, or the relevant ASTM/customer method) defines the target and the test. Always run the **exact method named in your job's spec**, and verify the stripping solution and acceptance limits against the supplier TDS and SDS. The specific standard/limits vary by customer — treat them as *verify-against-spec*, not as one fixed number.



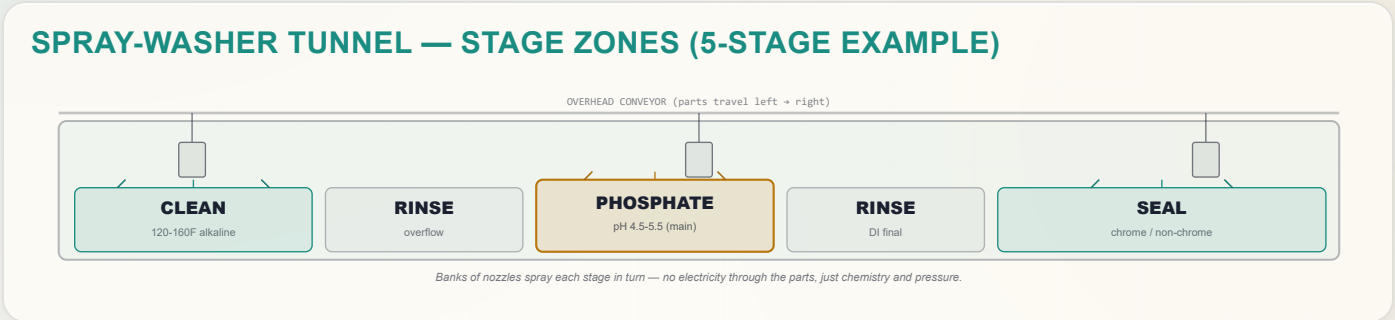
#### TIP

Run a **test panel** through the line with each significant load and log its color (and periodically its gravimetric weight). A simple coating-weight control chart catches a drifting bath *before* you've sent a shift's worth of bad parts to paint.

LINE CONFIGURATIONS

# IMMERSION VS. SPRAY • 3-STAGE VS. 5-STAGE

SAME CHEMISTRY FAMILY, DIFFERENT MACHINES AND DIFFERENT NUMBERS



|                      | SPRAY WASHER                          | IMMERSION (DIP)             |
|----------------------|---------------------------------------|-----------------------------|
| Coverage of recesses | Weaker (spray shadows; rotate/orient) | Stronger (floods cavities)  |
| Speed / throughput   | High, continuous (conveyor)           | Lower, batch                |
| Contact times        | Short (30–90 s phosphate)             | Longer (1–3 min phosphate)  |
| Mist / ventilation   | Higher concern                        | Lower                       |
| Typical use          | High-volume paint/powder lines        | Complex parts, lower volume |

## • 3-STAGE VS. 5-STAGE (AND BEYOND)

- **3-stage:** Cleaner-Coater → Rinse → Seal. Lowest cost and footprint. Lighter coatings, less tolerance for heavy soil. Great for lightly-soiled steel and budget lines.
- **5-stage:** Clean → Rinse → Phosphate → Rinse → Seal. Dedicated cleaner + dedicated phosphate = better cleaning, heavier/more uniform coatings, more robust. The workhorse.
- **6/7-stage and beyond:** add a **DI final rinse**, extra recirculated rinses, or a fresh-water “halo” rinse for top corrosion performance.

**TIP**

More stages = more control and better corrosion performance, at higher cost and footprint. The right choice is driven by the **corrosion spec and the parts’ soil load**, not by “more is always better.” When a 3-stage line struggles with heavily oiled parts or can’t hit the coating weight, that’s the classic signal it should have been a 5-stage.

— KNOW YOUR SEAL

# SEAL CHEMISTRY : CHROME VS. NON-CHROME

THE SINGLE MOST SERIOUS CHRONIC-HEALTH DECISION ON THE LINE

|                 | CHROMATE (CR(VI)) SEAL                       | NON-CHROME SEAL                     |
|-----------------|----------------------------------------------|-------------------------------------|
| Corrosion boost | Historically excellent                       | Comparable for most jobs            |
| Health hazard   | Confirmed carcinogen; OSHA 1910.1026 program | Much lower                          |
| Waste handling  | Chromium waste treatment required            | Easier                              |
| Trend           | Being phased out                             | Standard for new/converted lines    |
| Examples        | Chromic / chromic-phosphoric rinses          | Zirconium/titanium, polymer/organic |

**HEADS-UP**

The single most serious chronic-health hazard on a typical iron phosphate line is a **Cr(VI) seal**. If your line still runs one, the full OSHA hexavalent-chromium program applies (exposure monitoring, respirators, medical surveillance, housekeeping). The cleanest fix is conversion to a **non-chrome (zirconium) seal** — discuss with your chemistry supplier and EHS team. (Final chemistry decisions go through your QC/EHS process — this manual advises, it doesn't approve.)

**TECHNICAL STUFF**

A seal works by reacting with or coating the slightly porous, chemically “open” conversion film to close those pores and add a final corrosion barrier. Chromate seals deposited a chromium-rich film that historically gave outstanding salt-spray numbers; modern zirconium/titanium and polymer seals build an ultra-thin reactive film that matches them for most jobs without the carcinogen. The performance verdict is always the customer's corrosion spec — validate, don't assume.

**REMEMBER**

**Know which seal your shop runs — it changes your PPE, your ventilation, and your waste handling.** Chromate means the full Cr(VI) program. Non-chrome means ordinary line PPE plus the product's SDS. When in doubt, ask before you work the seal station.

— WHERE THE INDUSTRY IS HEADING

# THE MODERN ALTERNATIVE : ZIRCONIUM “NANO” THIN-FILM

**AN HONEST LOOK, BECAUSE YOU MAY SEE YOUR SHOP CONVERT**

The biggest shift in this field: **zirconium- (and titanium-) based thin-film pretreatments** — often marketed as “nano” or “thin-film” — are increasingly replacing iron phosphate.

## • WHY THEY'RE WINNING

- **Low/ambient temperature** — little or no heating (big energy savings vs. a heated phosphate stage).
- **Far less sludge** — they deposit an ultra-thin (nanometer-scale) zirconium-oxide film, so there's little iron-phosphate sludge to filter and dispose of.
- **No phosphate in the wastewater** — easing a regulated discharge.
- **Works on mixed metals** — steel, aluminum, and galvanized on the *same line*, where iron phosphate only really coats steel.
- **Fewer stages possible** — can simplify the line.

## • HONEST CAVEATS

- Tighter process control (the film is *extremely* thin), very clean incoming surfaces, and good rinse-water quality matter even more.
- Performance vs. iron/zinc phosphate is application-dependent — validate against the customer's corrosion spec.



### REMEMBER

Iron phosphate is mature, cheap, and well-understood — and still everywhere. But the industry trend is toward **zirconium thin-film** for energy, waste, and mixed-metal reasons. A well-rounded operator knows both, and knows *why* a shop might switch.



### FUN FACT

A zirconium film is measured in *nanometers* — billionths of a meter, tens to a few hundred atoms thick. An iron phosphate film, already thin at a fraction of a thousandth of an inch, is dozens of times thicker. The newest “coating” in the shop is one you essentially cannot see at all.

## THE QUIET CULPRITS

# WATER, RINSE & WASTE

## WHERE HIDDEN PROBLEMS — AND REAL LEGAL OBLIGATIONS — LIVE

### • WATER & RINSE QUALITY

- **Incoming water hardness** (calcium/magnesium) leaves spots and can interfere with coating and seal — many lines use **softened or DI water**, especially for the **final rinse**.
- **The last rinse before the seal/dry should be the cleanest** (ideally **DI**) — its residue ends up under the paint.
- **Monitor rinse conductivity** as your early-warning gauge for drag-in.

**TIP**

If you're chasing spots, streaks, or inconsistent paint adhesion and the chemistry checks out, **suspect the rinse water**. Hard or contaminated rinse water is a quiet, common culprit.

### • WASTE & THE ENVIRONMENT

Iron phosphating has real environmental obligations — know them:

- **Phosphate discharge** is regulated (it's a nutrient that causes algae blooms); rinse overflows and bath dumps need proper treatment/permitting.
- **Iron-phosphate sludge** must be filtered/settled, collected, and disposed of as the appropriate waste.
- **Spent acidic baths and alkaline cleaners** require neutralization/treatment before discharge.
- **Chromate (Cr(VI)) seal waste** is hazardous and requires chromium reduction/treatment — another reason shops move to non-chrome.
- **Oils skimmed** from the cleaner are a waste stream too.

**HEADS-UP**

Never dump any bath, rinse, or skimmed oil down a drain on your own judgment. Discharges are permitted and treated. **Follow your shop's waste procedures and your wastewater permit** — violations carry serious legal and environmental consequences.

— HOW WE PROVE IT WORKED

# QUALITY CHECKS

IRON PHOSPHATE IS JUDGED BY WHAT THE PAINT DOES, NOT BY THE BARE COATING

| CHECK                        | WHAT IT TELLS YOU                 | HOW                                                     |
|------------------------------|-----------------------------------|---------------------------------------------------------|
| Water-break test             | Is the part truly clean?          | Watch water sheet vs. bead (free, every load)           |
| Coating color                | Approx. coating weight/uniformity | Visual: gray / blue / gold / tan                        |
| Coating weight (gravimetric) | Exact weight vs. spec             | Weigh → strip → reweigh ÷ area                          |
| Paint adhesion (crosshatch)  | Does paint actually grip?         | <b>ASTM D3359</b> tape/crosshatch on painted samples    |
| Salt spray over paint        | Under-paint corrosion resistance  | <b>ASTM B117</b> salt-fog on painted samples (per spec) |



### REMEMBER

Iron phosphate is judged by what the *paint* does, not by the bare coating. The real verdicts come from **crosshatch adhesion (ASTM D3359)** and **salt spray over paint (ASTM B117)** on finished samples. Coating weight and color are the *in-process* predictors of those final results.



### TECHNICAL STUFF — WHAT IS “SALT SPRAY”?

It’s an accelerated corrosion test (per ASTM B117): painted samples sit in a sealed cabinet sprayed with salt fog, and you record how many *hours* until corrosion creeps from a scribe or rust/blisters appear. More hours = better protection. It’s how customers specify and verify corrosion requirements on the finished, painted part.



### TIP

The crosshatch (D3359) test is fast and shop-friendly: cut a small grid through the paint, press and pull tape, and see how much paint lifts. Clean grid, nothing lifts = good adhesion = your phosphate did its job. Lots of squares peeling = look upstream at cleaning and phosphate weight.

FIELD REFERENCE

TROUBLESHOOTING
QUICK-INDEX

FIND YOUR SYMPTOM, READ THE LIKELY CAUSE, TRY THE FIX

| SYMPTOM                           | LIKELY CAUSE(S)                                                            | FIRST THINGS TO CHECK                                              |
|-----------------------------------|----------------------------------------------------------------------------|--------------------------------------------------------------------|
| No coating / bare gray            | Dirty part; low accelerator; high free acid; low temp                      | Water-break test; accelerator & free-acid titration; bath temp     |
| Powdery / chalky / tan-brown      | Coating too heavy; free acid off; temp too high; bath imbalance            | Lower temp/time; adjust free acid per TDS; check total:free ratio  |
| Uneven / patchy coating           | Poor cleaning; spray shadows; nesting; bad nozzles                         | Re-clean upstream; reorient load; inspect/unclog nozzles           |
| Rust after coating / before paint | Flash rust from sitting wet; weak/skipped seal; incomplete dry-off         | Move parts promptly; verify seal; verify dry-off temp/time         |
| Paint blisters / peels later      | Phosphate salts under paint (rinse); trapped water (dry-off); fingerprints | Improve final (DI) rinse; verify dry-off; handle with clean gloves |
| Spots / streaks                   | Hard or dirty rinse water; drying water spots                              | Soften/DI rinse water; ensure clean final rinse                    |
| Coarse / rough film               | High sludge / bath contamination                                           | Filter or refresh bath; check incoming part condition              |
| Coating weight drifting           | Bath drifting (acid/accelerator/temp); concentration drift                 | Titrate and replenish per TDS; check temperature control           |



TIP

The fastest diagnostic instinct: **read the part backward through the line**. Bare gray? Suspect cleaning and phosphate. Blisters? Suspect rinse and dry-off. Powdery? Suspect phosphate weight/temperature. Rust-after? Suspect seal and dry-off. The defect almost always names the stage that failed it.



HEADS-UP

Before you change bath chemistry, temperature, or accelerator, confirm the change with your lead and the supplier TDS — and remember the accelerator is an **oxidizer**. Bath adjustments and additions are made only by authorized personnel, in full PPE, with ventilation running.

## — YOUR DECODER RING

# GLOSSARY — EVERY TERM, PLAIN AND SIMPLE

ANYTIME A WORD IN THIS MANUAL MADE YOU PAUSE, FLIP BACK HERE

**Accelerator:** An oxidizer added to the phosphate bath to speed and uniform the coating reaction. Low accelerator = thin or no coating. Handle as an oxidizer hazard.

**Amorphous:** Having no regular crystal structure. Iron phosphate coatings are amorphous (and thin); zinc/manganese phosphates are crystalline (and heavier).

**Ambient:** Room temperature; no heater.

**Cleaner-coater:** A single product that cleans *and* deposits the phosphate coating in one stage; enables 3-stage lines.

**Coating weight:** Mass of coating per unit area (mg/ft<sup>2</sup> or g/m<sup>2</sup>). The key iron-phosphate spec; ~30–90 mg/ft<sup>2</sup> typical.

**Conductivity:** How well water carries electricity; used to measure rinse cleanliness (µS/cm). Lower = cleaner.

**Confined space:** An enclosed space (like a washer tunnel) not designed for occupancy; entry requires a permit/procedure.

**Conversion coating:** A coating formed by chemically reacting (converting) the metal's own surface — no electric current.

**Cr(VI) / hexavalent chromium:** A confirmed human carcinogen used in older chromate seals; strict OSHA controls (29 CFR 1910.1026).

**DI water (deionized):** Water with minerals removed; rinses cleaner, fewer spots.

**Dried-in-place:** A seal applied with no rinse afterward; the part goes straight to dry-off.

**Drag-out / drag-in:** Solution clinging to a part leaving a tank (drag-out) and arriving in the next (drag-in) — the contamination rinses fight.

**Dry-off oven:** Oven that flashes off all water before paint (separate from the paint/powder cure oven).

**Free acid / total acid:** Titration “points” used to control the phosphate bath; their ratio indicates bath health.

**Iridescence:** The blue-to-gold color of a thin film (interference); used to read iron-phosphate coating weight.

**Iron phosphate:** A light, amorphous conversion coating (iron phosphate + iron oxides) used as a paint/powder base on steel.

**LOTO (Lockout/Tagout):** De-energizing and locking out equipment before maintenance.

**Manganese phosphate:** A heavy, black crystalline phosphate for wear/break-in/oil retention.

**Passivation / seal:** The final reactive rinse that closes the coating's pores and boosts under-paint corrosion resistance.

**pH:** Acidity/alkalinity scale (0 acid – 14 base; 7 neutral). Iron phosphate runs ~4.5–5.5.

**Pickling:** Acid dissolving of metal/oxide from a surface; the gentle pickling action is what starts the phosphate reaction.

**Pretreatment:** Surface preparation done *before* painting/powder; iron phosphate is a pretreatment.

**Spray washer:** Enclosed tunnel where conveyor-borne parts are sprayed through each stage.

**Sludge:** Iron-phosphate precipitate that builds up in the bath and must be filtered/disposed.

**Substrate:** The base metal being coated (here, steel/iron).

**TDS / SDS:** Technical Data Sheet (how to run a product) / Safety Data Sheet (its hazards). Always follow both.

**Water-break test:** Free cleanliness check: water sheets off a clean part, beads on an oily one.

**Zinc phosphate:** A heavier, gray crystalline phosphate; stronger paint/corrosion/lube base than iron phosphate.

**Zirconium pretreatment (“nano”/thin-film):** Modern low-temperature, low-sludge alternative to phosphating; works on mixed metals.

— TEMPLATE — PRINT ONE PER OPERATOR

# OPERATOR TRAINING MATRIX

HOW A SUPERVISOR PROVES — ON PAPER — THAT EACH OPERATOR IS TRAINED AND SIGNED OFF BEFORE RUNNING SOLO



REMEMBER

An operator is “line-qualified” only when every station they run is signed. Nobody works the **dry-off oven**, a **chromate seal**, or **enters a washer tunnel** solo until the **Safety / PPE & SDS** row is signed. Safety qualification comes first, always.

Sign-off levels: **O** = Observed only · **A** = Assisted under supervision · **Q** = Qualified to run solo · **T** = Qualified to Train others.

| #  | STATION / SKILL                                        | O/A/Q/T | TRAINER INIT & DATE | OPERATOR INIT & DATE | RE-CERT DUE |
|----|--------------------------------------------------------|---------|---------------------|----------------------|-------------|
| 1  | Safety / PPE & SDS (all stations)                      | _____   | _____               | _____                | _____       |
| 2  | Station 01 — Inspect & Load (drainage, substrate)      | _____   | _____               | _____                | _____       |
| 3  | Station 02 — Alkaline Clean + water-break test         | _____   | _____               | _____                | _____       |
| 4  | Station 03 — Rinse (post-clean) + conductivity         | _____   | _____               | _____                | _____       |
| 5  | Station 04 — Iron Phosphate / Cleaner-Coater + color   | _____   | _____               | _____                | _____       |
| 6  | Bath control — free/total acid & accelerator titration | _____   | _____               | _____                | _____       |
| 7  | Station 05 — Rinse (post-phosphate, DI final)          | _____   | _____               | _____                | _____       |
| 8  | Station 06 — Seal (identify chrome vs. non-chrome)     | _____   | _____               | _____                | _____       |
| 9  | Cr(VI) program (if chromate seal in use)               | _____   | _____               | _____                | _____       |
| 10 | Station 07 — Dry-Off Oven operation                    | _____   | _____               | _____                | _____       |
| 11 | Station 08 — Inspect & Hand-Off to Paint               | _____   | _____               | _____                | _____       |
| 12 | Coating-weight measurement (gravimetric)               | _____   | _____               | _____                | _____       |
| 13 | LOTO (pumps/heaters/conveyor/oven)                     | _____   | _____               | _____                | _____       |
| 14 | Confined-space awareness (washer tunnel)               | _____   | _____               | _____                | _____       |
| 15 | Emergency response — eyewash, shower, spill            | _____   | _____               | _____                | _____       |
| 16 | Waste / drag-out / sludge handling                     | _____   | _____               | _____                | _____       |
| 17 | Troubleshooting — symptom recognition                  | _____   | _____               | _____                | _____       |

Operator name: \_\_\_\_\_ Hire/start date: \_\_\_\_\_

Supervisor name: \_\_\_\_\_ Review cycle: ☐ Annual   ☐ Other: \_\_\_\_\_



HEADS-UP

Safety rows (1, 13, 14, 15) are not optional and not “fill in later.” An operator may not work *any* station until PPE/SDS, LOTO, confined-space awareness, and emergency response are signed. Safety competency comes before production competency, always.

PART FOUR — BACK MATTER & CERTIFICATION  
 20 QUESTIONS • PASSING SCORE 80% (16/20)

# IRON PHOSPHATE COMPLETION TEST

COVERS EVERY STATION PLUS SAFETY — ANSWER IN YOUR OWN WORDS WHERE ASKED



## REMEMBER

**Passing score: 80% (16 of 20 correct).** Any **safety** question answered wrong (Q5, Q6, Q15, Q19) is an automatic fail of the safety gate — re-teach and re-test before sign-off, regardless of total score. The Answer Key follows — no peeking until you're done!

Name: \_\_\_\_\_ Date: \_\_\_\_\_ Score: \_\_\_\_\_ / 20 **Safety-gate Qs (all correct): 5, 6, 15, 19**

**Q1.** Iron phosphate coating forms by:

- A. An electric current depositing iron from a bath onto the part B. A chemical reaction between the steel surface and the process solution (no electric current) C. Spraying molten phosphate onto the steel D. Baking a phosphate powder onto the part in the oven

**Q2. (Short answer)** What is the **water-break test**, and what result tells you a part is clean?

**Q3.** Typical iron phosphate **coating weight** is about:

- A. 500–1,000 mg/ft<sup>2</sup> B. 5–10 mg/ft<sup>2</sup> C. 30–90 mg/ft<sup>2</sup> (0.3–1.0 g/m<sup>2</sup>) D. 2,000 mg/ft<sup>2</sup>

**Q4. (Short answer)** Name iron phosphate's **main job** (what it's *for*). Is it a standalone corrosion finish? Explain in one sentence.

**Q5. [SAFETY GATE]** Which **seal** chemistry is a **confirmed human carcinogen** requiring a special OSHA program (29 CFR 1910.1026)?

- A. Hexavalent chromium (Cr(VI)) chromate seal B. Zirconium-based non-chrome seal C. Plain DI-water final rinse D. Organic/polymer seal

**Q6. [SAFETY GATE]** Before climbing into a spray-washer **tunnel** to clear nozzles or do maintenance, you must:

- A. Just hold your breath and be quick B. Turn the lights off C. Wait for the parts to cool D. Follow the confined-space procedure: LOTO, atmosphere testing, ventilation, attendant, and a permit

**Q7. (Short answer)** Why does the **final rinse before the seal/dry** ideally use clean or **DI water**? What does leftover residue cause under the paint?

**Q8.** The iron phosphate stage operates at what **pH**?

- A. 4.5–5.5 B. 1.0–2.0 C. 7.0 (neutral) D. 11–13

**Q9. (Short answer)** You can “read” iron phosphate coating weight by **color**. What do **gray**, **blue/gold**, and **tan/powdery** each tell you?

**Q10.** An iron phosphate coating that is **too heavy** typically looks:

- A. Bright mirror-like B. Deep glossy black C. Tan/brown and powdery (and gives *worse* paint adhesion) D. Bright blue and glassy

**Q11. (Short answer)** Name the **three** routine bath-control checks for an iron phosphate stage and, in a few words, what each tells you.

**Q12.** A **cleaner-coater** is best described as a product that:

- A. Only removes oil, nothing else   B. Cleans *and* deposits the phosphate coating in one stage (enabling a 3-stage line)   C. Only seals the coating  
D. Replaces the dry-off oven

**Q13.** The modern, **low-energy / low-sludge** alternative increasingly replacing iron phosphate (and good on mixed metals) is:

- A. Cyanide zinc plating   B. Hot-dip galvanizing   C. Manganese phosphate   D. Zirconium / titanium “nano” thin-film pretreatment

**Q14. (Short answer)** Iron phosphate works best on which **substrate**, and what happens if you run **aluminum or galvanized** parts through it?

**Q15. [SAFETY GATE]** The alkaline cleaner stage is hazardous mainly because it is:

- A. Radioactive   B. Hot and caustic — it causes chemical burns and its mist irritates the airways   C. Explosive on contact with air   D. Completely harmless

**Q16.** Why must parts be **fully dry** (dry-off oven) before paint/powder?

- A. To make them shiny   B. To cool them down   C. Trapped water boils out during cure and causes blisters/pops in the finish   D. It isn't necessary

**Q17. (Short answer)** In plain terms, how does iron phosphate form **without any electric current**? (Two or three sentences.)

**Q18.** Iron phosphate differs from **zinc** and **manganese** phosphate mainly because it is:

- A. The lightest and thinnest (amorphous), used as a light paint base   B. The heaviest of the three   C. Always black   D. An electroplated metal layer



**SAFETY SECTION — MUST BE CORRECT TO CERTIFY**

Questions 5, 6, 15, and 19 cover hazards that can permanently harm you or others. A miss on any safety item means re-teach and re-test before sign-off.

**Q19. (Short answer / Safety) [SAFETY GATE]** List **four** pieces of PPE you must wear at the **iron phosphate / alkaline cleaner** stations, and name **one** additional hazard control required because the **accelerator is an oxidizer**.

**Q20. (Short answer)** Name **two** honest **tradeoffs/weaknesses** of iron phosphate compared with zinc phosphate or zirconium pretreatment.

*End of test. Hand to your trainer for grading.*

FOR TRAINERS &amp; SELF-CHECKERS

# ANSWER KEY

SHORT-ANSWER RESPONSES DON'T NEED TO MATCH WORD-FOR-WORD — LOOK FOR THE KEY CONCEPTS IN BOLD

**HEADS-UP**

Keep this page out of the trainee's copy. Questions **5, 6, 15, and 19** are safety-critical — any wrong answer there requires re-training on that topic before sign-off, regardless of total score.

**Self-check answer distribution (the 11 multiple-choice items): A = 3, B = 3, C = 3, D = 2.** If your key clusters on one letter, you've mis-keyed.

**Q1. B** — A chemical reaction between the steel surface and the solution; **no electric current**.

**Q2.** Pull the part and watch the water: a **continuous, unbroken sheet** = clean; **beading** = oil remains, re-clean. (Free — use it every load.)

**Q3. C** — ~30–90 mg/ft<sup>2</sup> (0.3–1.0 g/m<sup>2</sup>); iron phosphate is a *light* coating.

**Q4.** It's a **paint/powder base (pretreatment)** — it improves paint adhesion and under-paint corrosion resistance on steel. **No, it is not a standalone corrosion finish** (that's zinc/manganese phosphate); bare, it offers only modest protection.

**Q5. A** — Hexavalent chromium (Cr(VI)) chromate seal (OSHA 29 CFR 1910.1026).

**Q6. D** — Follow the confined-space procedure: LOTO, atmosphere testing, ventilation, attendant, and permit.

**Q7.** Because residue left under paint is **hygroscopic phosphate salt** that attracts moisture and causes **blisters** (and poor adhesion). DI water leaves the least residue. (Credit for "residue/salts under paint cause blisters/adhesion failure.")

**Q8. A** — 4.5–5.5.

**Q9.** **Gray/no color** = too light (poor adhesion); **blue→gold iridescent** = in the target range (good paint base); **tan/brown/powdery** = too heavy (loose, chalky, *worse* adhesion).

**Q10. C** — Tan/brown and powdery; it gives worse adhesion, not better.

**Q11.** **Free acid** (unreacted acid — drives coating formation), **total acid** (total content; the total:free ratio shows bath health), and **accelerator** (the oxidizer that speeds/uniforms the coating; low = thin/no coating).

**Q12. B** — Cleans and coats in one stage; enables a 3-stage line.

**Q13. D** — Zirconium/titanium "nano" thin-film pretreatment.

**Q14.** Best on **steel/iron**. On **aluminum or galvanized**, little or no true iron-phosphate film forms (it mostly just cleans them) — flag it; a **zirconium** pretreatment is usually the right process for mixed metals.

**Q15. B** — Hot and caustic; causes chemical burns and its mist irritates the airways. (Required PPE + ventilation.)

**Q16. C** — Trapped water boils out during cure and blisters/pops the finish.

**Q17.** The **mildly acidic** solution lightly **pickles the steel** (iron dissolves, hydrogen forms); right at the surface the **local pH rises**, which makes dissolved iron and phosphate **precipitate** as an integral iron-phosphate/iron-oxide film. An **accelerator** drives it. No electricity needed — the metal and solution react on their own.

**Q18. A** — The lightest/thinnest (amorphous), used as a light paint base.

**Q19. PPE (any four):** sealed chemical splash goggles, face shield, chemical-resistant gloves, chemical apron, chemical-resistant/non-slip boots, respirator if ventilation is inadequate. **Oxidizer control (any one):** keep accelerator away from fuels/solvents/rags; don't let spills dry on combustibles; store/segregate per SDS; ensure ventilation/mist control.

**Q20.** Any two of: **lighter protection** than zinc/manganese phosphate (fails demanding corrosion specs); **steel-only** (poor on aluminum/galvanized); **makes sludge** (filtering/disposal); **phosphate in wastewater** (regulated discharge); **needs heat/energy** vs. ambient zirconium; **being displaced by zirconium** thin-film for energy/waste/mixed-metal reasons.

**REMEMBER**

16/20 passes — but every safety question (5, 6, 15, 19) must be correct to certify. Miss a safety item → re-teach and re-test before sign-off. We don't gamble with people.

PRINT, FILL IN, SIGN



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CERTIFICATE OF  
COMPLETION

IRON PHOSPHATE PRETREATMENT TRAINING PROGRAM

This certifies that

\_\_\_\_\_  
OPERATOR NAME

has successfully completed the **Iron Phosphate** training course — covering all eight process stations (Inspect/Load, Alkaline Clean, Rinse, Iron Phosphate/Cleaner-Coater, Rinse, Seal, Dry-Off, and Inspect/Hand-Off to Paint), the fundamentals of chemical conversion coating without electric current, coating-weight and color control, the cleaner-coater concept, seal chemistry (chrome vs. non-chrome/zirconium), the modern zirconium thin-film alternative, troubleshooting, and the safety practices for alkaline cleaners, acidic phosphate solutions, oxidizing accelerators, chromate seal chemistry, hot ovens, mist, and washer-tunnel confined spaces — and has passed the Completion Test with a score of \_\_\_\_\_ / 20, including all required safety competencies.

This operator is hereby cleared for **independent work** at the certified stations listed on the Operator Training Matrix.

Date of completion: \_\_\_\_\_ Re-certification due: \_\_\_\_\_

\_\_\_\_\_  
TRAINEE SIGNATURE

\_\_\_\_\_  
CERTIFYING TRAINER

\_\_\_\_\_  
SUPERVISOR

Training reference only. Verify all parameters against your chemistry supplier's TDS, customer specifications, and applicable regulatory requirements before production use. Hexavalent chromium, where used in a seal, is a confirmed human carcinogen — always consult current SDS and comply with OSHA, EPA, REACH, and local regulations.

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